BSc (Hons) Artificial Intelligence and Data Science

Module: CM4609 Individual Research Project

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Predictive Maintenance using Machine Learning and Time-Series Forecasting

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1 Introduction

The apparel manufacturing industry currently relies on reactive maintenance , leading to unexpected machine breakdowns and production downtime. While IoT devices on machines generate rich real-time data, current machine learning models are not utilizing this information effectively. This project aims to bridge the forecasting gap by developing a system that warns of breakdowns before they occur.

2 Research Objectives

The core research objectives focus on implementing a full-stack predictive maintenance solution

- **Data Collection:** Capture real-time IoT data from sewing machines.
- **Preprocessing:** Process vibration, voltage, current, and machine metadata.
- **Modeling:** Build a time-series forecasting model.
- **Prediction:** Forecast breakdowns ahead of time.
- **Insight Generation:** Provide interpretable insights into machine behavior.

3 Data Architecture & Pipeline

The data pipeline is designed to move from raw sensor input to actionable model outputs:

- **IoT Sensors:** Capture high-frequency sensor data to detect early signs of failure.
- **Microcontroller:** Processes and buffers data, adds timestamps, and handles secure transmission.
- **Azure IoT Hub:** Manages secure device authentication and data routing.
- **Azure SQL Database:** Stores structured, time-series records for analysis and training.
- **Python Preprocessing:** Cleans, normalizes, and engineers features from the raw data.
- **ML Model:** Detects failure patterns to predict issues and support maintenance.

4 Technical Analysis & Modeling

Analysis shows distinct patterns distinguishing healthy operations from pre-failure states. Healthy patterns are defined by stable, consistent signals , while breakdown patterns often show irregular, sharp spikes in vibration and fluctuations in electrical signals (voltage/current).

Challenges & Solutions

- **Dataset Imbalance:** Few breakdown samples exist compared to healthy ones, complicating training.

- **Solutions:** The project utilizes synthetic data generation to augment rare breakdown cases and anomaly-aware modeling (e.g., one-class models, cost-sensitive learning) to detect deviations.

The Temporal Fusion Transformer (TFT)

The TFT is selected as the primary modeling approach because it:

- Combines LSTM and attention layers to capture temporal patterns.
- Processes multiple sensor inputs simultaneously, weighting features based on their predictive power.
- Learns long-term and short-term dependencies.
- Provides interpretability via attention insights, helping teams trust predictions.
- Has proven success in industrial forecasting applications.

5 Current Implementation Status & Next Steps

Current Status

- **Data Ingestion:** Connected to Azure SQL with automated, secure pipelines.
- **Preprocessing:** Completed cleaning and normalization.
- **Feature Extraction:** Ready; includes time/frequency-domain features.
- **Modeling:** Baseline LSTM model is currently running and trained.

Next Steps

- **Benchmarking:** Train multiple models (LSTM, Prophet, TFT, Google Time FM) and compare performance using RMSE, MAE, and F1 metrics.
- **Validation:** Test against real factory failures to validate early warnings and lead-time accuracy.
- **Dashboarding:** Create an Early Warning System dashboard for maintenance teams.